(1)

a) $z \in R^k$, $y \in R^n$ so $\frac{\partial z}{\partial y}$ is $k \times n$, $x \in R^m$ so $\frac{\partial y}{\partial x}$ is $n \times m$

$\Rightarrow \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial x}$ is $k \times m$ matrix

$$\frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial x} = \begin{bmatrix} \frac{\partial z_1}{\partial y_1} & \frac{\partial z_1}{\partial y_2} & \cdots & \frac{\partial z_1}{\partial y_n} \\ \vdots & & & \vdots \\ \frac{\partial z_k}{\partial y_1} & \frac{\partial z_k}{\partial y_2} & \cdots & \frac{\partial z_k}{\partial y_n} \end{bmatrix} \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \cdots & \frac{\partial y_1}{\partial x_m} \\ \vdots & & & \\ \frac{\partial y_n}{\partial x_1} & \frac{\partial y_n}{\partial x_2} & \cdots & \frac{\partial y_n}{\partial x_m} \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{n} \frac{\partial z_1}{\partial y_i} \frac{\partial y_i}{\partial x_1} & \cdots & \sum_{i=1}^{n} \frac{\partial z_1}{\partial y_i} \frac{\partial y_i}{\partial x_m} \\ \vdots & & \vdots \\ \sum_{i=1}^{n} \frac{\partial z_k}{\partial y_i} \frac{\partial y_i}{\partial x_1} & & \sum_{i=1}^{n} \frac{\partial z_k}{\partial y_i} \frac{\partial y_i}{\partial x_m} \end{bmatrix} \quad (1)$$

$$\frac{\partial z}{\partial x} = \begin{bmatrix} \frac{\partial z_1}{\partial x_1} & \frac{\partial z_1}{\partial x_2} & \cdots & \frac{\partial z_1}{\partial x_m} \\ \vdots & & & \vdots \\ \frac{\partial z_k}{\partial x_1} & \frac{\partial z_k}{\partial x_2} & \cdots & \frac{\partial z_k}{\partial x_m} \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{n} \frac{\partial z_1}{\partial y_i} \frac{\partial y_i}{\partial x_1} & \cdots & \sum_{i=1}^{n} \frac{\partial z_1}{\partial y_i} \frac{\partial y_i}{\partial x_m} \\ \vdots & & \vdots \\ \sum_{i=1}^{n} \frac{\partial z_k}{\partial y_i} \frac{\partial y_i}{\partial x_1} & & \sum_{i=1}^{n} \frac{\partial z_k}{\partial y_i} \frac{\partial y_i}{\partial x_m} \end{bmatrix} \quad (2)$$

(1) , (2) $\Rightarrow \frac{\partial z}{\partial x} = \frac{\partial z}{\partial y} \frac{\partial y}{\partial x}$

b) $y = Ax$

$A : m \times n$, $x : n \times 1$

$$Ax = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{n1} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} , \quad \frac{\partial y}{\partial x} = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \cdots & \frac{\partial y_1}{\partial x_n} \\ \vdots & & & \vdots \\ \frac{\partial y_m}{\partial x_1} & \frac{\partial y_m}{\partial x_2} & \cdots & \frac{\partial y_m}{\partial x_n} \end{bmatrix}$$

$y_1 = a_{11} x_1 + a_{12} x_2 + \cdots a_{n1} x_n \Rightarrow \frac{\partial y_1}{\partial x_1} = a_{11}$, $\frac{\partial y_1}{\partial x_2} = a_{12}$, $\cdots \frac{\partial y_1}{\partial x_n} = a_{n1}$

Similarly, $y_m = a_{m1} x_1 + a_{m2} x_2 + \cdots + a_{mn} x_n \Rightarrow \frac{\partial y_m}{\partial x_1} = a_{m1}$, $\cdots$, $\frac{\partial y_m}{\partial x_n} = a_{mn}$

So $\frac{\partial y_i}{\partial x_j} = a_{ij} \Rightarrow \frac{\partial y}{\partial x} = A$

c) Since $y = f(x)$ is an element-wise function, if $i=j$ then $f(x_i) = y_i$

$\Rightarrow \frac{\partial y_i}{\partial x_i} = \frac{\partial}{\partial x_i} f(x_i) = f'(x_i)$ for $i = j$

When $i \neq j$, $\frac{\partial y_i}{\partial x_j} = 0$ since y_i does not depend on x_j

$\Rightarrow \frac{\partial y}{\partial x} = \text{diag}(f'(x))$

d) Let $H = Ug(Wx)$, $G = g(Wx)$, $Y = (Wx)$

We can write $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial H} \frac{\partial H}{\partial G} \frac{\partial G}{\partial Y} \frac{\partial Y}{\partial x}$

④ $\frac{\partial z}{\partial H}$

$z = f(Ug(Wx)) \Rightarrow z = f(H)$ where f is element wise

$\Rightarrow \frac{\partial z}{\partial H} = diag(f'(H)) = diag(f'(Ug(Wx)))$

⑤ $\frac{\partial H}{\partial G}$ where $H = UG$

$\Rightarrow \frac{\partial H}{\partial G} = U$

⑥ $\frac{\partial G}{\partial Y}$ where $G = g(Y)$, g is elementwise

$\Rightarrow \frac{\partial G}{\partial Y} = diag(g'(Y)) = diag(g'(Wx))$

⑦ $\frac{\partial Y}{\partial X}$ where $Y = Wx$

$\Rightarrow \frac{\partial Y}{\partial X} = W$

Therefore, $\frac{\partial z}{\partial x} = diag(f'(Ug(Wx))) \cdot U \cdot diag(g'(Wx)) \cdot W$

e) $\frac{\partial J}{\partial W} = \frac{\partial J}{\partial z} \frac{\partial z}{\partial W}$

$z = Wx$ so $z_i = W_{i1}x_1 + W_{i2}x_2 + \dots + W_{in}x_n$ (1)

$\frac{\partial J}{\partial W_{ij}} = \frac{\partial J}{\partial z_i} \frac{\partial z_i}{\partial W_{ij}}$ (2)

(1), (2) $\Rightarrow \frac{\partial z_i}{\partial W_{ij}} = x_j$

$$\Rightarrow \frac{\partial z}{\partial W} = \begin{bmatrix} \frac{\partial z}{\partial W_{11}} & \frac{\partial z}{\partial W_{21}} & \cdots & \frac{\partial z}{\partial W_{m1}} \\ \vdots & & & \\ \frac{\partial z}{\partial W_{1n}} & \frac{\partial z}{\partial W_{2n}} & \cdots & \frac{\partial z}{\partial W_{mn}} \end{bmatrix} = X$$

$\Rightarrow \frac{\partial J}{\partial W} = X \frac{\partial J}{\partial z}$

f) In matrix representation, $\hat{y} = softmax(z) = \frac{\exp(z)}{1^T \exp(z)}$

$\Rightarrow J = -y^T \log \hat{y} = -y^T \log\left(\frac{\exp(z)}{1^T \exp(z)}\right) = -y^T z + y^T \log(1^T \exp(z))$

$\frac{\partial J}{\partial z} = -y^T + y^T \frac{\partial}{\partial z} \log(1^T \exp(z))$

$$= -y^T + y^T \operatorname{diag}\left(\frac{1}{1^T \exp(z)}\right) \frac{\partial}{\partial z}\left(1 1^T \exp(z)\right) \quad \text{since log is elementwise}$$

$$= -y^T + y^T \operatorname{diag}\left(\frac{1}{1^T \exp(z)}\right) 1 1^T \frac{\partial}{\partial z} \exp(z)$$

$$= -y^T + y^T \operatorname{diag}\left(\frac{1}{1^T \exp(z)}\right) 1 1^T \operatorname{diag}(\exp(z)) \quad \text{since exp is elementwise}$$

$$= -y^T + y^T \operatorname{diag}\left(\frac{1}{1^T \exp(z)}\right) 1 \exp(z)^T$$

$$= -y^T + y^T \begin{bmatrix} \frac{\exp(z)^T}{1^T \exp(z)} \\ \vdots \\ \frac{\exp(z)^T}{1^T \exp(z)} \end{bmatrix} = -y^T + \operatorname{softmax}(z)^T = \operatorname{softmax}(z)^T - y^T = \hat{y}^T - y^T = (\hat{y} - y)^T$$

(2)

a) The problem with this frequency based estimator is that if w_j never appears in class k then $c(w_j, y=k) = 0$

$\Rightarrow P(w_j \mid y=k) = 0$

so according to posterior probability, if $P(w_j \mid y=k) = 0$ then $P(y=k \mid x) = 0$

b) i)

$$-\log P(w_j \mid y=k) = -\log\left(\frac{\exp(v_i^T c_k)}{\sum_{j \in V} \exp(v_j^T c_k)}\right) = -v_i^T c_k + \log\left(\sum_{j \in V} \exp(v_j^T c_k)\right)$$

① $$\frac{\partial}{\partial v_j} \log(\exp(v_j^T c_k)) = \operatorname{diag}\left(\frac{1}{\exp(v_j^T c_k)}\right) \frac{\partial}{\partial v_i} \exp(v_j^T c_k)$$

$$= \operatorname{diag}\left(\frac{1}{\exp(v_j^T c_k)}\right) c_k \cdot \operatorname{diag} \exp(v_j^T c_k)$$

$$= c_k \operatorname{softmax}(v_j \cdot c_k)^T$$

$$\Rightarrow \frac{\partial}{\partial V} \log\left(\sum_{j \in V} \exp(v_j^T c_k)\right) = c_k \operatorname{softmax}(V c_k)^T$$

② $-v_i^T c_k = -c_k 1_i^T V$ where 1_i^T is one hot vector with 1 at index i

$$\Rightarrow \frac{\partial}{\partial V} -v_i^T c_k = -c_k 1_i^T$$

$$\Rightarrow \frac{\partial}{\partial V} \text{Loss} = \sum_{(x,k) \in D} \sum_{w_j \in x} c_k \left(\operatorname{softmax}(V c_k) - 1_i\right)^T \quad \text{(since derivative of prior part} = 0\text{)}$$

ii) $$\frac{\partial}{\partial c_k}\left(-v_i^T c_k\right) = -v_i^T$$

$$\frac{\partial}{\partial c_k} \log\left(\sum_{j \in V} \exp(v_j^T c_k)\right) = \text{diag}\left(\frac{1}{\sum_{j \in V} \exp(v_j^T c_k)}\right) \frac{\partial}{\partial c_k} \sum_{j \in V} \exp(v_j^T c_k)$$

$$= \text{diag}\left(\frac{1}{\sum_{j \in V} \exp(v_j^T c_k)}\right) \text{diag}(\exp(V c_k)) \frac{\partial}{\partial c_k} \sum_{j \in V} (v_j^T c_k)$$

$$= \text{softmax}(V c_k)^T \frac{\partial}{\partial c_k} (V c_k) = \text{softmax}(V c_k)^T V$$

The prior part is not related to c_k so its derivative $= 0$

$$\Rightarrow \frac{\partial}{\partial c_k} \text{Loss} = -v_i^T + \text{softmax}(V c_k)^T V = -1_i^T V + \text{softmax}(V c_k) V$$

$$= (-1_i^T + \text{softmax}(V c_k)) V$$

iii) We can replace softmax with sigmoid function $\sigma(x) = \frac{1}{1+\exp(-x)}$

$$\text{Loss} = -\log \sigma(v_i^T c_k) - \sum_{j=1}^{M} \log \sigma(-v_{neg_j}^T c_k)$$

where the 1st term maximizes probability of positive words

the 2nd term minimizes probability of M negative sampling words.